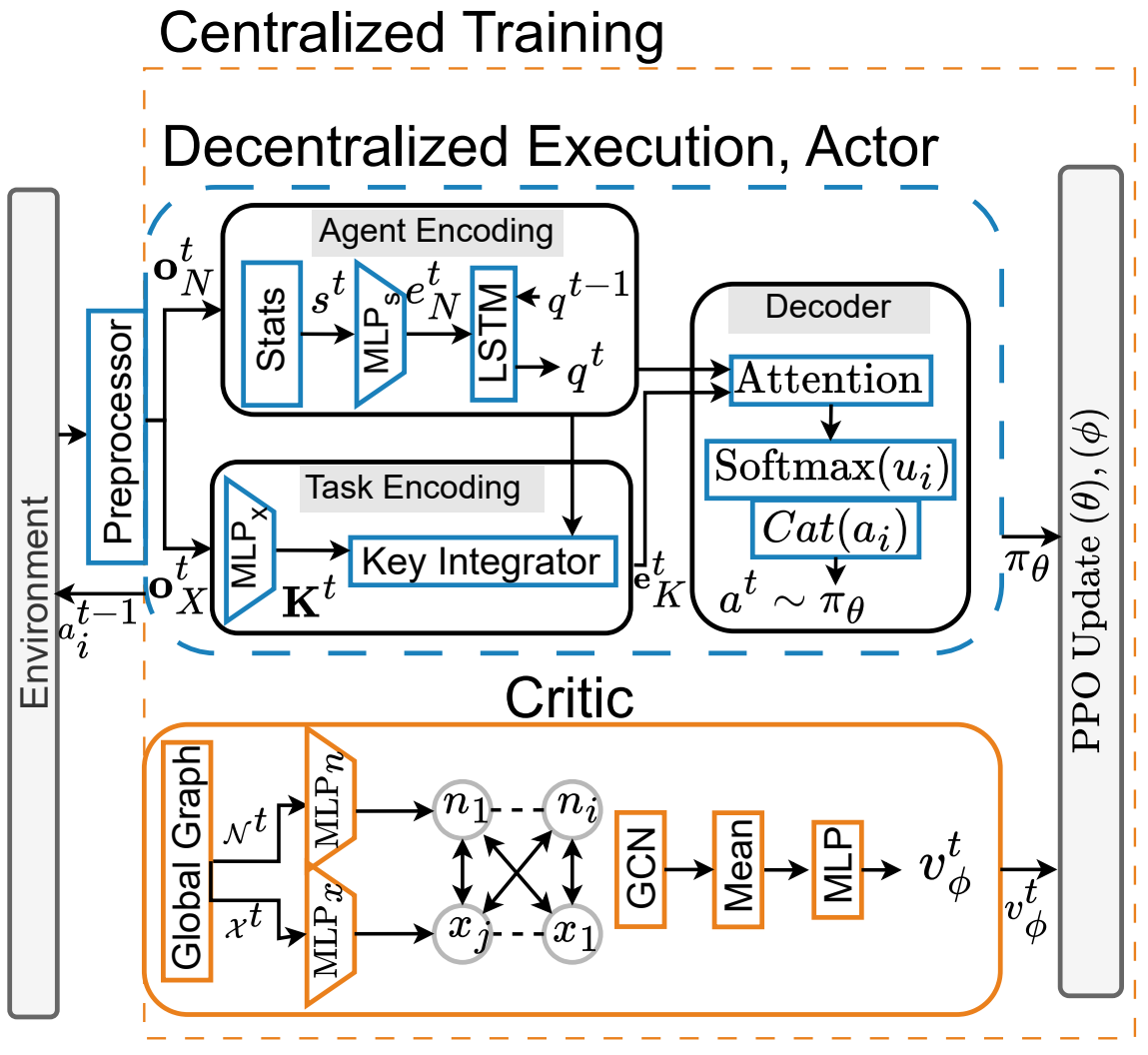

Centralized Training
Decentralized Execution, Actor
Environment
Preprocessor
$\mathbf{o}_N^t$
$\mathbf{o}_X^t$
a_i^{t-1}
Agent Encoding
Stats
s^t
MLP_s
e_N^t
LSTM
q^{t-1}
q^t
Task Encoding
MLP_x
$\mathbf{K}^t$
Key Integrator
$\mathbf{e}_K^t$
Decoder
Attention
$\mathrm{Softmax}(u_i)$
$Cat(a_i)$
$a^t \sim \pi_\theta$
π_θ
PPO Update $(\theta), (\phi)$
Critic
Global Graph
$\mathcal{N}^t$
x^t
MLP_n
MLP_x
n_1
n_i
x_j
x_1
GCN
Mean
MLP
v_ϕ^t
v_ϕ^t